



Lecture 12

Let's see an example problem
Suppose that you found a transfer function as follows

$$G(s) = \frac{(s+2)(s+4)}{s(s+1)(s+3)}$$

Find the time domain function $g(t)$ that corresponds to this transfer function.

$G(s)$ can be written as a partial fraction expansion

$$G(s) = \frac{C_1}{s} + \frac{C_2}{s+1} + \frac{C_3}{s+3}$$

using the cover up method, we get

$$C_1 = \frac{(s+2)(s+4)}{(s+1)(s+3)} \Big|_{s=0}$$

$$= \frac{8}{3}$$

$$C_2 = \frac{(s+2)(s+4)}{s(s+3)} \Big|_{s=-1}$$

$$= -\frac{3}{2}$$

$$C_3 = \frac{(s+2)(s+4)}{s(s+1)} \Big|_{s=-3}$$

$$= -\frac{1}{6}$$

$$G(s) = \frac{\frac{8}{3}}{s} + \frac{-\frac{3}{2}}{s+1} + \frac{-\frac{1}{6}}{s+3}$$

By taking inverse Laplace transform we get

$$g(t) = \frac{2}{3} - \frac{3}{2}e^{-t} - \frac{1}{6}e^{-3t}, \quad t \geq 0$$

Poles of the transfer function determines its stability. For a stable system, the poles should always lie in the left half plane in the s-domain.